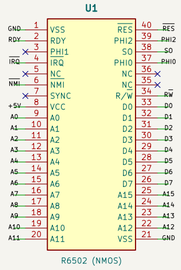
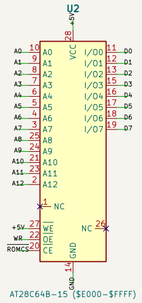
flappy6502 — R6502 + 28C64 + 32K SRAM + 74LS logic -> ST7789 LCD (bit-banged SPI)

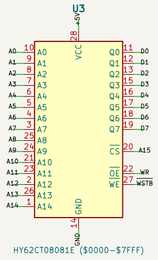
Memory map: RAM $0000–$7FFF, LCD latch $8000 (bit0 SDA, bit1 SCL, bit2 DC, bit3 CS), ROM $E000–$FFFF

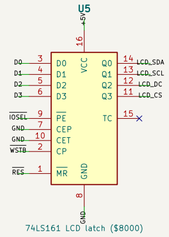
CPU clock PHI0 = U6 QC (1.25 MHz) or QD pin 11 (625 kHz). C4 sits at U5 pin 2.

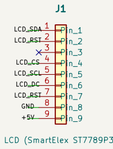
LCD rule: never feed the module 5 V; RST and EN come from U7 (low during reset).







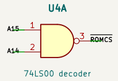


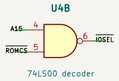


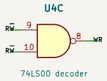


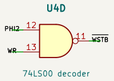


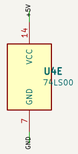
J1 pins: 1 SDA 2 RST 3 TE 4 CS 5 SCL 6 DC 7 EN 8 GND 9 VCC

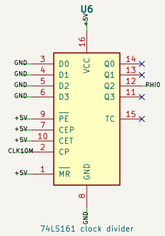












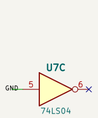


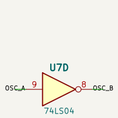


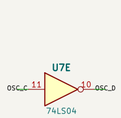


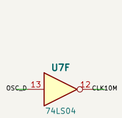










































Decoupling: C9–C15 100n at each IC (U1–U7); C8 10u at the CPU.















Final working build (bit-banged SPI, flappy_bb.bin)
Label-based nets: same label = same net

Sheet: /
File: flappy6502.kicad_sch

Title: flappy6502 — R6502 Flappy Bird breadboard computer

Size: A2	Date: 2026-10-27	**Rev: 1**
KiCad E.D.A. 10.0.5		Id: 1/1